

Intraosseous Radiofrequency Ablation of the Basivertebral Nerve for Chronic Vertebrogenic Low Back Pain

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Description of Procedure or Service

Intraosseous radiofrequency ablation of the basivertebral nerve is a minimally invasive, outpatient procedure for patients with vertebrogenic pain. Intraosseous radiofrequency ablation of the basivertebral nerve is intended to be used in conjunction with radiofrequency (RF) generators for the ablation of basivertebral nerves of the L3 through S1 vertebrae.

Vertebrogenic pain is caused when the vertebral endplates degenerate and cause pain due to the damage to intraosseous nerves. The intraosseous nerve (basivertebral) weaves through the vertebral bodies. This can result in chronic low back pain.

The diagnosis of vertebral endplate pain is made using a classification scale known as Modic changes (Type 1 or Type 2). Modic changes are areas of bone marrow damage that appear on Magnetic Resonance Imaging (MRI).

Benefit Application

This medical policy relates only to the services or supplies described herein. Please refer to the Member's Benefit Booklet for availability of benefits.

Policy Statement

GEHA will provide coverage for surgical treatment for intraosseous radiofrequency ablation of the basivertebral nerve when it is determined to be medically necessary because the medical criteria and guidelines as documented below have been demonstrated.

When treatment for thermal destruction of the intraosseous basivertebral nerve is covered:

Thermal destruction of the intraosseous basivertebral nerve will be considered medically reasonable and necessary for the treatment of chronic lumbar back pain in patients who meet **ALL** the following criteria:

1. Chronic lumbar back pain of ≥ 6 months duration that causes functional deficit measured on a pain or disability scale*, AND
2. Documented failure to respond to ≥ 6 months of non-surgical management**, AND
3. Absence of non-vertebrogenic pathology per clinical assessment or radiology studies that could explain the source of the patient's pain, including but not limited to fracture, tumor, infection, or significant deformity, AND
4. Evidence of Type 1 or Type 2 Modic changes on MRI, such as inflammation, edema, vertebral endplate changes, disruption and fissuring of the endplate, vascularized fibrous tissues within the adjacent marrow, hypotensive signals (Type 1 Modic change), and changes to the vertebral body marrow including replacement of normal bone marrow by fat, and hypertensive signals (Type 2 Modic change), in 1 or more vertebrae from L3-S1.

NOTE: Thermal destruction of the intraosseous BVN must only be performed once per vertebral body from L3-S1 per lifetime. Up to 4 vertebral bodies may be treated during 1 procedure.

* Pain assessment and a disability scale must be obtained at baseline to be used for functional assessment.

** Non-surgical management may include but is not limited to:

- Avoidance of activities that aggravate pain;
- Trial of Chiropractic manipulation;
- Trial of Physical Therapy;
- Cognitive support and recovery reassurance;
- Injection therapy – epidural and/or facet;
- Spine biomechanics education;
- Specific lumbar exercise program;
- Home use of heat/cold modalities;

- Low impact aerobic exercise as tolerated;
- Pharmacotherapy (e.g., non-narcotic analgesics, NSAIDs, muscle relaxants, neuroleptics, and narcotics).

When treatment for thermal destruction of the intraosseous basivertebral nerve is not covered:

Thermal destruction of the intraosseous basivertebral is considered not medically necessary or unproven for the following:

1. Skeletally immature (≤ 18 years old);
2. Severe cardiac or pulmonary compromise;
3. Active systemic infection or local infection at the intended treatment level;
4. Bleeding diathesis;
5. Pregnancy;
6. Primary radicular pain into the lower extremities (defined as nerve pain following a dermatomal distribution and that correlates with nerve compression on imaging);
7. Previous lumbar/lumbosacral spine surgery at the intended treatment level (with the exception of discectomy/laminectomy if performed >6 months prior to BVN nerve ablation and radicular pain resolved);
8. Primary symptomatic lumbar or lumbosacral spinal stenosis (defined as the presence of neurogenic claudication and confirmed by imaging);
9. Diagnosed osteoporosis (T-score of -2.5 or less), spine fragility fracture history, trauma/compression fracture at the intended treatment level, or spinal cancer;
10. Radiographic evidence of any of the following that correlates with predominant physical complaints:
 - a) Lumbar/lumbosacral disc extrusion or protrusion >5 mm at levels L3-S1;
 - b) Lumbar/lumbosacral spondylolisthesis > 2 mm at any level;
 - c) Lumbar/lumbosacral spondylolysis at levels L3-S1;
 - d) Lumbar/lumbosacral facet arthrosis/effusion correlated with facet-mediated pain at levels L3-S1.
11. Advanced generalized systemic disease that limits quality-of-life (QOL) improvements would require a statement of the objective of treatment in such cases;
12. Active, untreated substance abuse disorder.

Required Documentation

The scales used for measurement of pain and/or disability must be documented in the medical record. Acceptable scales include but are not limited to verbal rating scales, Numerical Rating Scale (NRS) and Visual Analog Scale (VAS) for pain assessment, and Pain Disability Assessment Scale (PDAS), Oswestry Disability Index (ODI), Oswestry Low Back Pain Disability Questionnaire (OSW), Quebec Back Pain

Disability Scare (QUE), Roland Morris Pain Scale, Back Pain Functional Scale (BPFS), and the PROMIS profile domains to assess function.

The patient's medical record should include but is not limited to:

- The assessment of the patient by the performing provider as it relates to the complaint of the patient for that visit,
- Relevant medical history (including a history of chronic vertebrologic LBP for at least 6 months and failure of non-surgical management of at least 6 months),
- Results of pertinent tests/procedures (including pertinent radiological studies, which must include MRI images and radiology reports of evidence of Modic changes Type 1 or 2 in the L3-S1 vertebral body endplates),
- Signed and dated office visit record/operative report.

Policy Guidelines

There are 2 types of Modic changes found on Magnetic Resonance Imaging (MRI):

Type 1 – Vascular development in the vertebral body, inflammation and edema, vertebral endplate changes, vascularized fibrous tissues within the adjacent marrow, hypointensive signals.

Type 2 – Changes in the vertebral body's bone marrow including replacement of normal bone marrow by fat, and hyperintensive signals.

American Society of Interventional Pain Physicians (ASIPP)

(ASIPP) ASIPP clinical practice guidelines reviewed the evidence for facet joint interventions for managing chronic spinal pain. The guidelines make the following recommendations:

- The level of evidence is II with moderate strength of recommendation for cervical and lumbar RFA.
- The level of evidence is III with weak to moderate strength of recommendation with emerging evidence for thoracic RFA.
- For facet joint nerve ablation, the suggested frequency would be 6 months or longer (maximum of 2 times per year) between each procedure, provided that 50% or greater relief is obtained for 5-6 months.
- If the interventional procedures are applied for different regions, they may be performed at intervals of no sooner than one week or preferably 2 weeks for most types of procedures if they are not allowed to be performed in one setting or contraindicated.
- The therapeutic frequency for medial branch neurotomy should remain at intervals of at least 6 months per region with multiple regions involved.

It is further suggested that all regions be treated at the same time, provided all procedures are performed safely.

- In the treatment or therapeutic phase, the interventional procedures should be repeated only as necessary according to the medical necessity criteria.

Society of Regional Anesthesia (ASRA) Pain Medicine Consensus

practice guidelines on interventions for lumbar facet joint pain from a multispecialty, international working group make the following recommendations:

- Medial branch blocks should be the prognostic screening test of choice before lumbar facet RFA.
- Repeat RFA procedures for recurrence of pain are recommended in patients who experienced a good outcome from the first RFA procedure, typically defined as at least 50% relief of pain at 3 months.
- Given the drop-off in success rates reported in some studies and the mean duration of benefit, the guidelines recommend repeating the procedure no more than two times per year.

National Institute for Health and Care Excellence (NICE) NICE guidelines (2016; updated 2020) on the management of LBP and sciatica make the following recommendations:

- Consider referral for assessment for radiofrequency denervation for people with CLBP when:
 - o Non-surgical treatment has not worked for them; and
 - o The main source of pain is thought to come from structures supplied by the medial branch nerve; and
 - o They have moderate or severe levels of localized back pain (rated as 5 or more on a visual analog scale, or equivalent) at the time of referral.
- Only perform radiofrequency denervation in people with CLBP after a positive response to a diagnostic medial branch block.
- Do not offer imaging for people with LBP with specific facet joint pain as a prerequisite for radiofrequency denervation.

North American Spine Society (NASS) NASS clinical guidelines provide evidence-based recommendations which are endorsed by the American Academy of Physical Medicine and Rehabilitation (AAPM&R) and American Association of Neurological Surgeons and Congress of Neurological Surgeons (AANS/CNS), for the diagnosis and treatment of adults with LBP. The guidelines make the following recommendations regarding RFA:

- Thermal RFA is suggested as a treatment for patients with LBP from the zygapophyseal joints. The outcomes of this procedure become more reliable when more stringent diagnostic criteria are used. The relief from these ablations is durable for at least six months following the procedure. Grade of recommendation: B - fair evidence (Level II or III studies with consistent findings) for or against recommending intervention.
- Cooled RFA of the sacral lateral branch nerves and dorsal ramus of L5 may be considered in patients with sacroiliac joint pain diagnosed with dual diagnostic blocks. Grade of recommendation: C - poor quality evidence (Level IV or V studies) for or against recommending intervention.
- There is insufficient evidence to make a recommendation for or against the use of cryodenervation for the treatment of zygapophyseal joint pain. Grade of recommendation: I - insufficient or conflicting evidence not allowing a recommendation for or against intervention.

International Society for the Advancement of Spine Surgery (ISASS)

The utilization of intraosseous BVN ablation to address vertebrogenic LBP has become a recognized safe, predictable, and durable surgical method for the management of chronic axial low back pain (LBP) identified using well-established clinical and magnetic resonance imaging findings, Modic type 1 and/or type 2 changes.

Intraosseous ablation of the BVN from the L3 through S1 vertebrae may be considered medically indicated for individuals with CLBP when all the following criteria are met:

Chronic low back pain (CLBP) of at least 6 months duration.

Failure to respond to at least 6 months of nonsurgical management.

[MRI]-demonstrated* MC1 or MC2 in at least 1 vertebral endplate at 1 or more levels from L3 to S1.

(*Endplate changes, inflammation, edema, disruption, and/or fissuring.)

Fibrovascular bone marrow changes (hypointense signal for Modic type 1).

Fatty bone marrow changes (hyperintense signal for Modic type 2)." (p. 7)

BVNA is NOT indicated in the following:

Patients with severe cardiac or pulmonary compromise.

Presence of implanted pulse generator(s) (e.g., pacemaker and defibrillator)/electronic implants except for circumstances where a specific patient safety precaution may be implemented.

Co-existence of other obvious radiographic etiology for patient's axial CLBP requiring a medically necessary surgical intervention.

Active or chronic infection, systemic or local.

Patients who are pregnant.

Skeletally immature patients (generally age <18 years).

Current or post-trauma, tumor, infection, or poor bone quality compromising vertebral pedicle/body.

Cauda equina syndrome defined as neural compression causing neurogenic bowel or bladder dysfunction.

Radiographic confirmation of gross spinal instability including angular or translatory instability (grade 2 or greater spondylolisthesis) at index level(s).

Morbid obesity precluding satisfactory procedural imaging.

Targeted ablation zone is <10 mm away from a sensitive structure not intended for ablation.

Situation where unintended tissue damage may result based on the clinical assessment by the physician.

The Centers for Medicare and Medicaid Services

LCD: L39420 Thermal Destruction of the Intraosseous Basivertebral Nerve (BVN) for Vertebrogenic Lower Back Pain

Approved: March 5, 2023

Covered Indications

Thermal destruction of the intraosseous BVN will be considered medically reasonable and necessary for the treatment of cLBP in patients who meet ALL the following criteria:

1. Chronic lumbar back pain of ≥ 6 months duration that causes functional deficit measured on a pain or disability scale*, AND
2. Documented failure to respond to ≥ 6 months of non-surgical management**, AND

3. Absence of non-vertebrogenic pathology per clinical assessment or radiology studies that could explain the source of the patient's pain, including but not limited to fracture, tumor, infection, or significant deformity, AND
4. Evidence of Type 1 or Type 2 Modic changes on MRI, such as inflammation, edema, vertebral endplate changes, disruption and fissuring of the endplate, vascularized fibrous tissues within the adjacent marrow, hypotensive signals (Type 1 Modic change), and changes to the vertebral body marrow including replacement of normal bone marrow by fat, and hypertensive signals (Type 2 Modic change), in 1 or more vertebrae from L3-S1.

* Pain assessment and a disability scale must be obtained at baseline to be used for functional assessment.

** Non-surgical management may include but is not limited to:

- Avoidance of activities that aggravate pain;
- Trial of Chiropractic manipulation;
- Trial of Physical Therapy;
- Cognitive support and recovery reassurance;
- Injection therapy – epidural and/or facet;
- Spine biomechanics education;
- Specific lumbar exercise program;
- Home use of heat/cold modalities;
- Low impact aerobic exercise as tolerated;
- Pharmacotherapy (e.g., non-narcotic analgesics, NSAIDs, muscle relaxants, neuroleptics, and narcotics).

Patients must have undergone careful screening, evaluation, and diagnosis by a multidisciplinary team prior to thermal destruction of the intraosseous BVN (such screening must include psychological, as well as physical evaluation). Documentation of the history and careful screening must be available in the patient chart if requested.

Limitations

Services that are not reasonable and necessary cannot be covered by Medicare in the following:

1. Skeletally immature patients (≤ 18 years old);
2. Severe cardiac or pulmonary compromise;
3. Active systemic infection or local infection at the intended treatment level;
4. Bleeding diathesis;
5. Pregnancy;
6. Primary radicular pain into the lower extremities (defined as nerve pain following a dermatomal distribution and that correlates with nerve compression on imaging);
7. Previous lumbar/lumbosacral spine surgery at the intended treatment level (with the exception of discectomy/laminectomy if performed >6 months prior to BVN nerve ablation and radicular pain resolved);

8. Primary symptomatic lumbar or lumbosacral spinal stenosis (defined as the presence of neurogenic claudication and confirmed by imaging);
9. Diagnosed osteoporosis (T-score of -2.5 or less), spine fragility fracture history, trauma/compression fracture at the intended treatment level, or spinal cancer;
10. Radiographic evidence of any of the following that correlates with predominant physical complaints:
 - a. Lumbar/lumbosacral disc extrusion or protrusion >5mm at levels L3-S1;
 - b. Lumbar/lumbosacral spondylolisthesis > 2mm at any level;
 - c. Lumbar/lumbosacral spondylolysis at levels L3-S1;
 - d. Lumbar/lumbosacral facet arthrosis/effusion correlated with facet-mediated pain at levels L3-S1.
11. BMI >40;
12. Advanced generalized systemic disease that limits quality-of-life (QOL) improvements would require a statement of the objective of treatment in such cases;
13. Active, untreated substance abuse disorder.

NOTE: Thermal destruction of the intraosseous BVN must only be performed once per vertebral body from L3-S1 per lifetime. Up to 4 vertebral bodies may be treated during 1 procedure.

Background

Low back pain is a leading cause of disability. Chronic low back pain (CLBP) is a persistent pain in the lumbar region lasting for longer than 3-6 months. Vertebrogenic CLBP is thought to be a distinct type of back pain, originating from degeneration of the vertebral body or vertebral body endplates, resulting in inflammation. The inflammatory response is perceived by the basivertebral nerve (BVN), a sensory nerve that enters the posterior vertebral body and branches out to the superior and inferior endplates. Pain signals are then transmitted to the central nervous system, causing vertebrogenic pain (Fischgrund et al., 2019).

Diagnosis of CLBP and its underlying cause is initiated with a comprehensive history and physical examination. Testing may include bloodwork, standard radiographs, bone scans, computerized axial tomography scans, MRI, electromyography, or a diagnostic injection (NASS, 2022).

CLBP can be treated with physical therapy (e.g., ice, heat, massage, transcutaneous electrical nerve stimulation [TENS], ultrasound), analgesia, anti-inflammatory medications, muscle-relaxing agents, coping skills, minimally invasive injections or procedures (e.g., nerve ablation), surgery, or complementary medicine (e.g., acupuncture, dry needling, magnets, etc.) (NINDS, 2022). Intraosseous BVN ablation using a radiofrequency probe and generator has been suggested as a minimally invasive treatment for CLBP in a subset of patients with specific clinical indications.

Fischgrund et. al (2018) conducted a randomized double blind sham controlled multi-center study to evaluate the safety and efficacy of radiofrequency (RF) ablation of the basivertebral nerve (BVN) for the treatment of chronic low back pain (CLBP) in a Food and Drug Administration approved Investigational Device Exemption trial. A total of 225 patients diagnosed with CLBP were randomized to either a sham (78 patients) or treatment (147 patients) intervention. The mean age within the study was 47 years (range 25-69) and the mean baseline ODI was 42. All patients had Type I or Type II Modic changes of the treated vertebral bodies. Patients were evaluated preoperatively, and at 2 weeks, 6 weeks and 3, 6 and 12 months postoperatively. The primary endpoint was the comparative change in ODI from baseline to 3 months. At 3 months, the average ODI in the treatment arm decreased 20.5 points, as compared to a 15.2 point decrease in the sham arm ($p = 0.019$, per-protocol population). A responder analysis based on ODI decrease ≥ 10 points showed that 75.6% of patients in the treatment arm as compared to 55.3% in the sham control arm exhibited a clinically meaningful improvement at 3 months. Conclusion: Patients treated with RF ablation of the BVN for CLBP exhibited significantly greater improvement in ODI at 3 months and a higher responder rate than sham treated controls. BVN ablation represents a potential minimally invasive treatment for the relief of chronic low back pain.

In a follow up study by Fischgurd et. al (2020), the long-term outcomes following intraosseous basivertebral nerve ablation for the treatment of chronic low back pain were assessed: 5-year treatment arm results from a prospective randomized double-blind sham-controlled multi-center study. In total, 117 US patients were treated successfully with BVN ablation. Patient-reported outcomes of post ablation treatments, and patient satisfaction were collected at a minimum of 5-years following BVN ablation. Comparisons between the post ablation and baseline values were made using an analysis of covariance with alpha 0.05.

Results: Of the 117 US treated patients 100 (85%) were available for review with a mean follow-up of 6.4 years (5.4-7.8 years). Mean Oswestry Disability Index (ODI) score improved from 42.81 to 16.86 at 5-year follow-up, a reduction of 25.95 points ($p < 0.001$). Mean reduction in Visual Analogue Scale (VAS) pain score was 4.38 points (baseline of 6.74, $p < 0.001$). In total, 66% of patients reported a $> 50\%$ reduction in pain, 47% reported a $> 75\%$ reduction in pain, and 34% of patients reported complete pain resolution. Composite responder rate using thresholds of ≥ 15 -point ODI and ≥ 2 -point VAS for function and pain at 5 years was 75%.

Conclusion: CLBP patients treated with BVN ablation exhibit sustained clinical improvements in function and pain with high responder rates at a mean of 6.4 years following treatment. BVN ablation is a durable, minimally invasive treatment for vertebrogenic CLBP.

In a prospective, randomized, multicenter study of intraosseous basivertebral nerve ablation for the treatment of chronic low back pain conducted by Khalil et. al (2019), comparison of the effectiveness of intraosseous RF ablation of the BVN to standard care for the treatment of chronic LBP in a specific subgroup of patients suspected to have vertebrogenic related symptomatology was evaluated. A total of

140 patients with CLBP of at least 6 months duration, with Modic Type 1 or 2 vertebral endplate changes between L3 and S1, were randomized 1:1 to undergo either Radiofrequency (RF) ablation of the BVN or continue standard care. Oswestry Disability Index (ODI) was collected at baseline, 3, 6, 9, and 12-months post procedure. Secondary outcome measures included a 10-point Visual Analog Scale (VAS) for LBP, ODI and VAS responder rates, SF-36, and EQ-5D-5L. The primary endpoint was a between-arm comparison of the mean change in ODI from baseline to 3 months post-treatment.

The interim analysis showed clear statistical superiority ($p < .001$) for all primary and secondary patient-reported outcome measures in the RF ablation arm compared with the standard care arm. This resulted in a data management committee recommendation to halt enrollment in the study and offer early cross-over to the control arm. These results are comprised of the outcomes of the 104 patients included in the intent-to-treat analysis of the 3-month primary endpoint, which included 51 patients in the RF ablation arm and 53 patients in the standard care arm. Baseline ODI was 46.1, VAS was 6.67, and mean age was 50 years. The percentage of patients with LBP symptoms ≥ 5 years was 67.3%. Comparing the RF ablation arm to the standard care arm, the mean changes in ODI at 3 months were -25.3 points versus -4.4 points, respectively, resulting in an adjusted difference of 20.9 points ($p < .001$). Mean changes in VAS were -3.46 versus -1.02, respectively, an adjusted difference of 2.44 cm ($p < .001$). In the RF ablation arm, 74.5% of patients achieved a ≥ 10 -point improvement in ODI, compared with 32.7% in the standard care arm ($p < 0.001$).

CONCLUSIONS: Minimally invasive RF ablation of the BVN led to significant improvement of pain and function at 3-months in patients with chronic vertebrogenic related LBP.

A Prospective, randomized, multicenter study of intraosseous basivertebral nerve ablation for the treatment of chronic low back pain: 24-Month treatment arm results was published October 2021 (Koreckij et.al). Results: 140 patients were randomized, 66 to BVNA. In the 58 BVNA patients completing a 24-month visit, 67% had back pain for > 5 years, 36% were actively taking opioids at baseline, 50% had prior epidural steroid injections, and 12% had prior low back surgery. Improvements in ODI, VAS, SF-36 PCS, and EQ-5D-5L were statistically significant at all timepoints through 2 years. At 24 months, ODI and VAS improved 28.5 ± 16.2 points (from baseline 44.5; $p < 0.001$) and 4.1 ± 2.7 cm (from baseline 6.6; $p < 0.001$), respectively. A combined responder rate of $ODI \geq 15$ and $VAS \geq 2$ was 73.7%. A $\geq 50\%$ reduction in pain was reported in 72.4% of patients and 31.0% were pain-free at 2 years. At 24 months, only 3(5%) of patients had BVNA-level steroid injections, and 62% fewer patients were actively taking opioids. There were no serious device or device-procedure related adverse events reported through 24 months.

In conclusion, Intraosseous BVNA demonstrates an excellent safety profile and significant improvements in pain, function, and quality of life that are sustained through 24 months in patients with chronic vertebrogenic low back pain.

Regulatory

Intraosseous radiofrequency ablation of the basivertebral nerve is a surgical procedure and thus not regulated by the FDA.

K190504 Trade/Device Name: Intracept Intraosseous Nerve Ablation System (RF Probe),
Intracept Intraosseous Nerve Ablation System (Access Instruments),
Relievent RF Generator Regulation Number: 21 CFR 882.4725
Regulation Name: Radiofrequency Lesion Probe Regulatory Class: Class II Product Code: GXI Dated:
February 28, 2019 Received: March 1, 2019
501 K premarket approval: May 2019

Intracept Intraosseous Nerve Ablation System Relievent Medsystems RF Generator (RFG) indications:

The Intracept Intraosseous Nerve Ablation System is intended to be used in conjunction with radiofrequency (RF) generators for the ablation of basivertebral nerves of the L3 through S1 vertebrae for the relief of chronic low back pain of at least six months duration that has not responded to at least six months of conservative care, and is also accompanied by features consistent with Type 1 or Type 2 Modic changes on an MRI such as inflammation, edema, vertebral endplate changes, disruption and fissuring of the endplate, vascularized fibrous tissues within the adjacent marrow, hypointensive signals (Type 1 Modic change), and changes to the vertebral body marrow including replacement of normal bone marrow by fat, and hyperintensive signals (Type 2 Modic change).

The Relievent RFG is intended to be used with RF probes FDA cleared as part of the Relievent Intracept Intraosseous Nerve Ablation System for the ablation of basivertebral nerves of the L3 through S1 vertebrae for the relief of chronic low back pain of at least six months duration that has not responded to at least six months of conservative care, and is also accompanied by features consistent with Type 1 or Type 2 Modic changes on an MRI such as inflammation, edema, vertebral endplate changes, disruption and fissuring of the endplate, vascularized fibrous tissues within the adjacent marrow, hypointensive signals (Type 1 Modic change), and changes to the vertebral body marrow including replacement of normal bone marrow by fat, and hyperintensive signals (Type 2 Modic change).

K250213 Trade/Device Name: OptaBlate BVN Intraosseous Nerve Ablation System (10mm Probe Single Kit);

OptaBlate BVN Intraosseous Nerve Ablation System (10mm Probe Dual Kit)

Date Received 01/24/2025

Decision Date 05/15/2025

501 K premarket approval: May 2025

The Stryker OptaBlate BVN Intraosseous Nerve Ablation System indications:

The Stryker OptaBlate BVN Intraosseous Nerve Ablation System is intended for the ablation of basivertebral nerves of the L3 through S1 vertebrae for the relief of chronic low back pain of at least six months duration that has not responded to at least six months of conservative care, and is also accompanied by features consistent with Type 1 or Type 2 Modic changes on an MRI such as inflammation, edema, vertebral endplate changes, disruption and fissuring of the endplate, vascularized fibrous tissues within the adjacent marrow, hypointensive signals (Type 1 Modic change), and changes

to the vertebral body marrow including replacement of normal bone marrow by fat, and hyperintensive signals (Type 2 Modic change).

Applicable Codes

The following list(s) of procedure and/or diagnosis codes is provided for reference purposes only and may not be all inclusive. Listing of a code in this policy does not imply that the service described by the code is a covered or non-covered health service. Benefit coverage for health services is determined by the member specific benefit plan document and applicable laws that may require coverage for a specific service. The inclusion of a code does not imply any right to reimbursement or guarantee claim payment. Other Policies and Guidelines may apply.

CPT/HCPCS Codes	Description
64628	Thermal destruction of intraosseous basivertebral nerve, including all imaging guidance; first 2 vertebral bodies, lumbar or sacral
64629	Thermal destruction of intraosseous basivertebral nerve, including all imaging guidance; each additional vertebral body, lumbar or sacral (List separately in addition to code for primary procedure)

Scientific References

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Policy History/Revision Information

October 2023	Origination
November 2024	Removed limiting criteria H. BMI>40; Updated document for better accessibility
January 1 2026	Removed deleted codes C9752 C9753. Removed code 22899
May 1 2026	Added information relating to newly approved intraosseous basivertebral nerve ablation devices: Policy Guidelines; Background and Regulatory. Added supporting scientific references.